Why Not Put a Microphone Near the Loudspeaker? A New Approach to Acoustic Echo Cancellation *

Fei Zhao[1,2], Xueliang Zhang[1], and Zhong-Qiu Wang[2]

[1] Inner Mongolia University, Hohhot, China
zhaofei@mail.imu.edu.cn, cszxl@imu.edu.cn
[2] Southern University of Science and Technology, Shenzhen, China
wang.zhongqiu41@gmail.com

Abstract. Acoustic echo cancellation (AEC) remains challenging due to loudspeaker nonlinearities and complex acoustics. We propose a dual-microphone configuration with a near-loudspeaker auxiliary reference microphone that captures the nonlinearly distorted far-end signal. To mitigate double-talk contamination, we purify the auxiliary reference using Wiener filtering with a compressed time-frequency mask to suppress near-end components. This purified reference signal enables a more effective linear AEC stage, whose residual error signal is then fed to a deep neural network for joint residual echo and noise suppression. Evaluation results show that our method outperforms baseline approaches on matched test sets. To evaluate its robustness under simulated strong unseen nonlinearities, we further test it on a mismatched dataset and observe substantial performance gains. These results demonstrate its effectiveness under simulated nonlinear conditions where the distortion type is unknown.

Keywords: Acoustic echo cancellation · Auxiliary reference microphone · Short-time Wiener filtering.

1 Introduction

Acoustic echo cancellation (AEC) suppresses echoes caused by acoustic coupling between loudspeakers and microphones in voice communication systems [1, 2]. Conventional AEC mainly uses adaptive filtering (e.g., NLMS, PFDAF) [3–5], but linear models struggle with nonlinear distortions from low-cost hardware (e.g., harmonic artifacts, clipping) [6, 7], often leaving residual echo in practice.

To address these limitations, deep learning-based AEC methods can be broadly grouped into three paradigms. **1) End-to-End DNNs** directly map microphone

* This work was done while Fei Zhao was a visiting student at SUSTech. This research was supported by National Key Research and Development Program of China (Grant No. 2025YFF0518003). *Corresponding author: Zhong-Qiu Wang.*

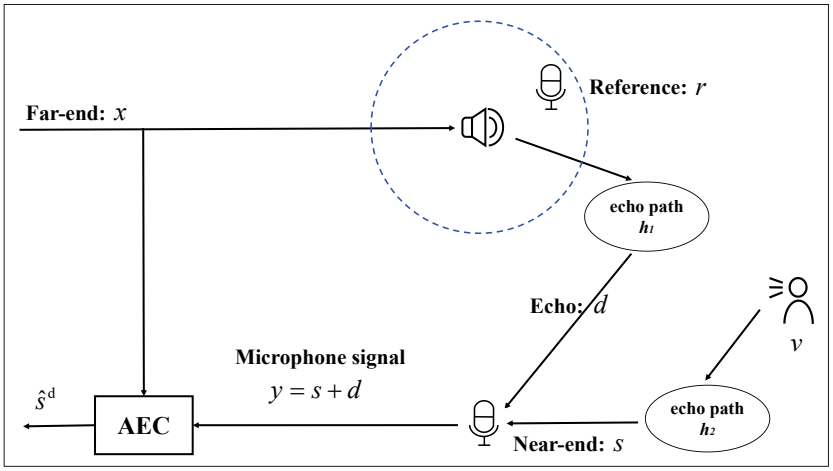


Fig. 1: Illustration of AEC aided by using an auxiliary reference microphone.

signals to echo-suppressed outputs [8–11], but their robustness to unseen environments and nonlinearities is often limited [12–14]. **2) Cascaded traditional–neural systems** combine adaptive filtering for coarse suppression with DNN-based residual removal [15–18], yet generalization can still be a bottleneck. **3) Neural–adaptive hybrid methods** use DNNs to predict adaptive-filter parameters [19–21]; they inherit some classical robustness but may sacrifice absolute suppression [20].

However, a fundamental difficulty in nonlinear AEC is *observability*: the far-end reference is not equal to the loudspeaker's *radiated* signal after nonlinear distortion. This reference mismatch makes the nonlinear echo hard to model and is a key factor behind poor *cross-nonlinearity generalization* under unseen distortions.

Nonlinear distortions in AEC often stem from loudspeaker/amplifier nonlinearities (e.g., hard clipping, saturation), causing harmonic and inter-modulation distortion [22]. To address loudspeaker-induced nonlinearity, we augment the conventional single-reference setup with a near-loudspeaker auxiliary reference microphone (Fig. 1) that measures the far-end signal *after* loudspeaker nonlinearity, providing a nonlinear reference unavailable with only the far-end signal. Although prior work has explored multi-microphone AEC [10], it typically relies on compact arrays placed away from the loudspeaker for spatial filtering; in contrast, our near-loudspeaker measurement is designed to capture nonlinear distortions. Under double-talk, we purify the auxiliary reference via short-time Wiener filtering and compressed IRM. Experimental results show robust performance under strong, unseen nonlinear distortions (e.g., clipping and saturation).

Contributions. This paper introduces a *sensing-driven* formulation of nonlinear AEC, where a new measurement improves nonlinearity observability and, in turn, generalization to unseen distortions.

- **New sensing channel for nonlinear observability and cross-nonlinearity generalization:** place an auxiliary reference microphone near the loudspeaker to capture the *radiated* far-end signal after loudspeaker nonlinearity, thereby improving the observability of the nonlinear echo source and enabling more robust AEC under mismatched nonlinear distortions;
- **Reference purification under double talk:** suppress near-end contamination in the auxiliary reference via short-time Wiener filtering followed by compressed IRM, yielding a far-end-dominant nonlinear reference;
- **Hybrid processing pipeline and targeted evaluation:** integrate WSTWS-based linear pre-cancellation with an ICCRN back-end, and validate the design on matched/mismatched nonlinear settings with ablations.

2 Problem Formulation

Fig. 1 shows our single-channel AEC system with an auxiliary reference microphone placed near the loudspeaker to capture its nonlinear output. Assuming weak ambient noise, the main-microphone signal y is modeled as a mixture of acoustic echo d and near-end speech s. The physical model in the time domain can be expressed as

$$y = x_{\mathrm{nl}} * h_1 + v * h_1 \tag{1}$$
$$= s + d \tag{2}$$
$$= s^{\mathrm{d}} + s^{\mathrm{r}} + d \in \mathbb{R}^N, \tag{3}$$

where $*$ denotes linear convolution and N is the signal length in samples. The near-end signal $s = v * h_1$ (with v denoting the anechoic near-end source) is decomposed into a direct-path component s^{d} and a late-reverberant component s^{r}. The acoustic echo d is generated by convolving the loudspeaker nonlinear output x_{nl} (a distorted version of the far-end signal x) with the echo-path RIR h_2.

The mixture signal captured by the auxiliary reference microphone, r, is a composite of the far-end signal component, r^{f}, and the contaminating near-end signal, r^{n}, formulated as:

$$r = x_{\mathrm{nl}} * h_3 + v * h_4 \tag{4}$$
$$= r^{\mathrm{n}} + r^{\mathrm{f}}, \tag{5}$$

where h_3 and h_4 denote the RIRs from the loudspeaker and near-end talker to the reference microphone, respectively. Due to the near-loudspeaker placement, the far-end component typically has much larger energy than the near-end component, which we exploit to isolate the nonlinear far-end signal for AEC.

3 Proposed method

The key advantage of our approach does not lie in a particular network architecture, but in the improved reference consistency enabled by the additional

sensing channel. Our proposed system, shown in Fig. 2, operates in the short-time Fourier transform (STFT) domain. It comprises two stages: a linear AEC stage, which suppresses the near-end component in the reference signal and performs signal processing-based linear AEC, and a neural AEC stage suppresses residual echoes.

3.1 Linear AEC Stage

Weighted short-time Wiener solution For the linear AEC component, we adopt the weighted short-time Wiener solution (WSTWS) [23], an enhancement over the conventional short-time Wiener solution (STWS) [24] by adaptively re-weighting the contribution of individual time-frequency (T-F) units [23]. The linear AEC result is obtained as

$$\mathcal{F}_{\text{WSTWS}}(Y, X) = Y(t, f) - \hat{\mathbf{h}}_1(t, f)^{\text{H}}\mathbf{X}(t, f). \tag{6}$$

which represents the result of canceling the estimated echo signal from the near-end mixture signal Y based on the far-end signal X. In Eq. (6), $\hat{\mathbf{h}}_1(t, f) \in \mathbb{C}^K$ denotes a K-tap linear filter modeling the echo path, and $\mathbf{X}(t, f) = [X(t, f), X(t-1, f), ..., X(t-K+1, f)]^{\mathsf{T}} \in \mathbb{C}^K$. In frame-online, real-time AEC, at the current frame t, WSTWS estimates the filter by solving the following problem:

$$\hat{\mathbf{h}}_1(t, f) = \underset{\mathbf{h_1}(f)}{\text{argmin}} \sum_{t'=t-W}^{t} \frac{\left|Y(t', f) - \mathbf{h}_1(f)^{\mathsf{H}}\mathbf{X}(t', f)\right|^2}{\lambda(t, f)}, \tag{7}$$

which is computed based on the current and the past W frames. The weighting term λ is computed as

$$\lambda(t, f) = \varepsilon \times \max_{t' \in [t-W, t]} \left(|Y(t', f)|^2\right) + |Y(t, f)|^2, \tag{8}$$

where ε is a floor value to avoid placing too much weight on low-energy T-F units.

Reference signal preprocessing Although the reference signal is dominated by far-end energy, it also contains near-end speech. To suppress this contamination, we first use WSTWS to estimate the near-end component and then apply a real-valued T-F mask [25]. Here, $\mathcal{F}_{\text{WSTWS}}(R, X)$ approximates the near-end contamination remaining in the reference microphone. Thus, $R - \mathcal{F}_{\text{WSTWS}}(R, X)$ approximates the far-end-dominant reference component used to form the mask. In detail,

$$\hat{M} = \frac{|R - \mathcal{F}_{\text{WSTWS}}(R, X)|}{|R - \mathcal{F}_{\text{WSTWS}}(R, X)| + |\mathcal{F}_{\text{WSTWS}}(R, X)|}, \tag{9}$$

$$R_m = \hat{M}^m \odot R, \tag{10}$$

where $\mathcal{F}_{\text{WSTWS}}(R, X)$ is the WSTWS result calculated based on the reference signal and the far-end signal, $m \in [0, 1]$ is a tunable mask compression factor,

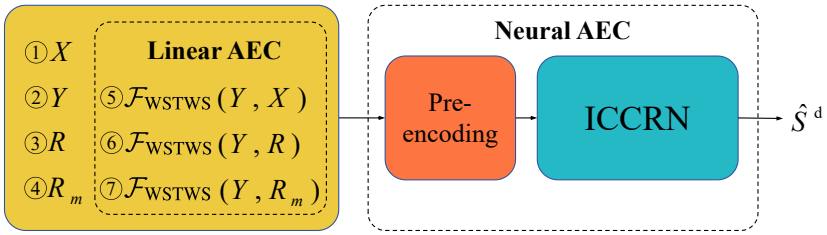


Fig. 2: Architecture of proposed AEC system.

and $\odot$ denotes point-wise multiplication. Because the far-end component in the reference signal has dominant energy, Eq. (9) preserves this dominant far-end component. With R_m, we can compute an additional linear AEC result, denoted as $\mathcal{F}_{\text{WSTWS}}(Y, R_m)$, besides using the reference signal R to directly compute a linear AEC result (i.e., $\mathcal{F}_{\text{WSTWS}}(Y, R)$).

3.2 Neural AEC Stage

Fig. 2 illustrates the overall architecture of the proposed AEC model. The network takes 7 input signals: ① far-end signal X, ② near-end mixture signal Y, and ③ reference signal R and ④ its masked version R_m. Additionally, we include the outputs of WSTWS applied to three signal pairs: ⑤ $\mathcal{F}_{\text{WSTWS}}(Y, X)$, ⑥ $\mathcal{F}_{\text{WSTWS}}(Y, R)$, and ⑦ $\mathcal{F}_{\text{WSTWS}}(Y, R_m)$, which respectively represent the linear AEC results between the far-end signal, the reference signal, the masked reference signal, and the near-end mixture signal. This multi-input design enables the model to leverage complementary echo estimates to improve AEC performance.

To harmonize the heterogeneous input representations and align their temporal dynamics, we apply a dedicated pre-encoding block to the real and imaginary components of each of the seven input signals. Each pre-encoding block consists of a causal 2D convolutional layer with a 3×3 kernel, 2 input channels and 2 output channels, followed by layer normalization [26] and PReLU [27]. The processed features are then concatenated along the channel dimension, resulting in a 14-channel input tensor to the AEC model. The AEC model employs the ICCRN architecture [28]. It is trained to output an estimate of the near-end speech signal, denoted as $\hat{S}^d$.

3.3 Loss Functions

The loss function combines the two loss functions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{RI+Mag}} + \alpha \times \mathcal{L}_{\text{S-SISNR}}, \tag{11}$$

where α is tuned to 0.01. The first loss term, $\mathcal{L}_{\text{RI+Mag}}$, penalizes the complex spectrum as follows:

$$\mathcal{L}_{\text{RI+Mag}} = \mathcal{L}_{\text{RI}} + \mathcal{L}_{\text{Mag}}, \tag{12}$$

$$\mathcal{L}_{\text{RI}} = \sum_{t,f} \left| |S^{\text{d}}(t,f)|^p e^{j\angle S^{\text{d}}(t,f)} - |\hat{S}^{\text{d}}(t,f)|^p e^{j\angle \hat{S}^{\text{d}}(t,f)} \right|^2, \tag{13}$$

$$\mathcal{L}_{\text{Mag}} = \sum_{t,f} \left| |S^{\text{d}}(t,f)|^p - |\hat{S}^{\text{d}}(t,f)|^p \right|^2, \tag{14}$$

where p (tuned to 0.5) is a spectral compression factor following [29]. The second loss term, $\mathcal{L}_{\text{S-SISNR}}$, denotes the stretched scale-invariant signal-to-noise ratio (S-SISNR) loss [30], which is a variant of SISNR [31]. It is a time domain loss function obtained by doubling the period of SISNR:

$$\mathcal{L}_{\text{S-SISNR}} = 10\log_{10} cot^2(\frac{\beta}{2}) = 10\log_{10} \frac{1+cos(\beta)}{1-cos(\beta)}, \tag{15}$$

where $cos(\beta)$ denotes the cosine similarity between the target speech signal s^d and predicted speech signal $\hat{s}^d$.

4 Experimental Setup

4.1 Datasets

The near-end and far-end signals are taken from the synthetic ICASSP 2023 AEC challenge dataset [16]. To the best of our knowledge, no public real-recording dataset supports AEC with a near-loudspeaker auxiliary reference microphone; collecting such data with controllable nonlinear playback requires dedicated hardware and measurement protocols and is beyond the scope of this study. We therefore evaluate the proposed method under simulated strong unseen nonlinearities. RIRs are generated with the image method [32]. Room dimensions $l \times w \times h$ are sampled with $l \in [4, 8]$ m, $w \in [3, 7]$ m, and $h \in [3, 5]$ m, and source positions (loudspeaker and near-end talker) are randomized. The auxiliary reference microphone is placed within a hollow sphere of radius 0.05–0.2 m centered at the loudspeaker, while the main microphone is sampled within $[l/10, l-1/10] \times [w/10, w-w/10] \times [1, \min(h-1, 3)]$. Reverberation time (T60) is sampled from 0.1 to 0.8 s. We generate 22,000 RIRs (20,000/1,000/1,000 for train/val/test) and 100,000/5,000/1,000 mixtures, where the echo is mixed with near-end speech s at SER in $[-10, 10]$ dB (step 1 dB). All mixtures are 6 s long at 16 kHz.

4.2 Nonlinearity Scenarios

This study considers five types of nonlinear settings. The first three [33] are used for training. For testing, we construct two test conditions: a matched condition, which employs the first three nonlinear settings, and a mismatched condition, which employs the last two, to verify the effectiveness of introducing auxiliary

reference microphones.

$$x_{nl}(n) = \frac{ax(n)}{\sqrt{a^2 + x^2(n)}}, \text{with } a = \frac{5}{b}, \tag{16}$$

$$x_{nl}(n) = 1 - e^{-ax(n)}, \text{with } a = \frac{b}{10}, \tag{17}$$

$$x_{nl}(n) = 2ax(n) + ax^2(n) + x^3(n), \text{with } a = \log\left(\frac{b}{10}\right) + 0.1, \tag{18}$$

where $b \in [2, 5]$. The other two nonlinearities are constructed by cascading clipping with sigmoid distortion. Specifically, the far-end signal is first processed by hard clipping [34] or soft clipping [35], and the clipped output is then passed through the sigmoid nonlinearity in Eq. (21).

$$x_{\text{hard}}(n) = \begin{cases} -x_{\max}, & x(n) < -x_{\max} \\ x(n), & |x(n)| \le x_{\max} \\ x_{\max}, & x(n) > x_{\max} \end{cases} \tag{19}$$

$$x_{\text{soft}}(n) = \frac{x(n)x_{\max}}{\sqrt{|x_{\max}|^\rho + |x(n)|^\rho}}, \tag{20}$$

where the clipping threshold $x_{\max}$ is set to 0.7 in this work. For soft-clipping saturation distortion, ρ is set to 2, denoting the non-adaptive shape parameter.

$$x_{\text{sigmoid}}(n) = \lambda\left(\frac{1}{1 + e^{-\nu \cdot z(n)}} - \frac{1}{2}\right), \tag{21}$$

where $z(n) = 1.5 \times x(n) - 0.3 \times x^2(n)$, and $x(n)$ denotes the clipped output $x_{\text{hard}}(n)$ or $x_{\text{soft}}(n)$ in the mismatched settings. The parameter λ denotes the sigmoid gain and is set to 2, and ν represents the sigmoid slope and is defined as

$$\nu = \begin{cases} 4, & \text{if } z(k) > 0 \\ 0.5, & \text{if } z(k) \le 0 \end{cases} \tag{22}$$

For the unmatched test set, the same sound source signals as in the matched test set are used, but with unseen nonlinearities.

4.3 Training Details

For STFT, the window size is 20 ms, hop size 10 ms, and the Hamming window is used as the analysis window. The model is trained using the Adam optimizer, starting with an initial learning rate of 0.001, which is halved if the validation loss does not improve for 2 epochs. Early stopping is applied if no improvement is observed for 8 epochs. The number of filter taps in WSTWS is set to $K = 20$ by default. For the reference preprocessing in Eq. (9), we set $K = 1$ because the loudspeaker and the reference microphone are spatially close. The flooring term ε in Eq. (8) is set to 0.001. The window size W in Eq. (7) is set to 200 frames. The mask compression factor m in Eq. (10) is set to 1/6.

4.4 Evaluation Setup

We consider three evaluation scenarios, including double talk (DT), near-end single talk (ST_NE), and far-end single talk (ST_FE). Three popular evaluation metrics are used, including echo return loss enhancement (ERLE) [36] for measuring echo suppression during far-end single talk, perceptual evaluation of speech quality (PESQ) [37] for assessing near-end speech quality, and signal-to-distortion ratio (SDR) [38] for quantifying near-end speech fidelity during double talk. In addition, EchoMOS from AECMOS [39] is added as one of the double-talk evaluation metrics; since the AECMOS evaluation model was trained on a real-recorded blind test set and may not accurately reflect the true performance of the proposed method, it is reported here only for reference. We follow the open-source AECMOS implementation, using the model *Run*_1663915512_*Stage*_0.*onnx* for evaluation. The evaluated models include standalone linear AEC methods, ICCRN, and ICCRN variants with different input features. The linear AEC baselines include PFDAF(Y, X) and PFDAF(Y, R), together with STWS and WSTWS evaluated on three signal pairs: (Y, X), (Y, R), and (Y, R_m). The two PFDAF settings use the same near-end mixture Y but different reference inputs: PFDAF(Y, X) uses the far-end signal X, whereas PFDAF(Y, R) uses the auxiliary reference-microphone signal R. For PFDAF, we set the number of blocks, FFT length, and step size to 200, 320, and 0.2, respectively, and use the optimized filter for initialization. The ICCRN variants are defined as follows:

- **ICCRN** uses only the far-end signal and the near-end mixture signal as inputs to predict the target speech.
- **ICCRN$_{\mathbf{r}}$**, which builds on ICCRN, additionally incorporates the reference signal into its network input.
- **ICCRN$_{\mathbf{rl}}$**, building on **ICCRN$_{\mathbf{r}}$**, further takes the linear AEC result computed based on the reference signal and the near-end mixture signal as input. It should be noted that the linear AEC result used here is computed based on STWS. That is, $\mathcal{F}_{\text{STWS}}(Y, R)$.
- **ICCRN$_{\mathbf{rl+fl}}$**, building on **ICCRN$_{\mathbf{rl}}$**, further incorporates the STWS result of the near-end mixture signal and the far-end signal. That is, $\mathcal{F}_{\text{STWS}}(Y, X)$.
- **ICCRN$_{\mathbf{rl+fl+mrl}}$**, building on **ICCRN$_{\mathbf{rl+fl}}$**, additionally includes the masked reference signal R_m in Section 3.1 and $\mathcal{F}_{\text{STWS}}(Y, R_m)$ to the input signal.
- **ICCRN$_{\mathbf{rl+fl+mrl+w}}$**, building on **ICCRN$_{\mathbf{rl+fl+mrl}}$**, replaces all the STWS operations with WSTWS. The detailed input signals are listed in Section 3.2.

Notice that ICCRN is considered the baseline of the other models, as this is currently the popular way to realize neural AEC.

5 Evaluation Results

Tables 1 and 2 evaluate the linear AEC baselines and the neural post-processing variants on matched and mismatched test sets. We first compare the standalone linear methods to isolate the effect of the reference signal. We then examine

Table 1: AEC results on the matched test set.

Test scenarios	DT			ST_NE		ST_FE
Model	PESQ	SDR (dB)	EchoMOS	PESQ	SDR (dB)	ERLE (dB)
Mixture	1.56	−11.8	-	2.25	−36.1	-
PFDAF(Y, X)	1.64	−9.0	2.79	-	-	2.6
PFDAF(Y, R)	1.67	−8.9	2.83	-	-	2.7
$\mathcal{F}_{\text{STWS}}(Y, X)$	1.77	−7.1	4.23	-	-	16.1
$\mathcal{F}_{\text{WSTWS}}(Y, X)$	1.85	−8.2	3.41	-	-	12.4
$\mathcal{F}_{\text{STWS}}(Y, R)$	1.76	−6.7	4.28	-	-	17.3
$\mathcal{F}_{\text{WSTWS}}(Y, R)$	1.88	−7.9	3.38	-	-	12.4
$\mathcal{F}_{\text{STWS}}(Y, R_m)$	1.76	−7.0	4.28	-	-	17.5
$\mathcal{F}_{\text{WSTWS}}(Y, R_m)$	1.88	−8.0	3.45	-	-	13.4
ICCRN	2.34	4.1	4.53	2.92	5.1	73.5
ICCRN_{r}	2.37	4.1	4.61	2.93	5.3	74.8
ICCRN_{rl}	2.49	4.7	4.61	2.96	5.1	73.4
$\text{ICCRN}_{\text{rl+fl}}$	2.52	4.7	**4.62**	2.93	5.3	76.1
$\text{ICCRN}_{\text{rl+fl+mrl}}$	2.55	5.0	4.60	2.95	5.8	77.2
$\text{ICCRN}_{\text{rl+fl+mrl+w}}$	**2.59**	**5.1**	4.60	**2.97**	**5.9**	**77.8**

whether the ICCRN back-end can further use these linear estimates to improve residual echo suppression and near-end speech quality.

On the matched test set (Table 1), replacing the far-end input X with the auxiliary reference R improves the PFDAF baseline from 1.64 to 1.67 in double-talk PESQ, from −9.03 to −8.90 dB in double-talk SDR, and from 2.61 to 2.74 dB in far-end single-talk ERLE. The same input advantage is clearer for the STWS family. Using R or R_m instead of X increases STWS ERLE from 16.12 dB to 17.32 and 17.54 dB, respectively. For WSTWS, the auxiliary-reference inputs also improve double-talk PESQ from 1.85 to 1.88, while R_m gives the highest WSTWS ERLE of 13.42 dB. Compared with STWS, WSTWS mainly improves perceptual quality in the standalone linear stage rather than uniformly improving all metrics. These comparisons show that the near-loudspeaker reference provides a more informative input for linear echo estimation than the far-end signal.

This trend becomes more pronounced on the mismatched test set (Table 2). For PFDAF, changing the reference input from X to R improves double-talk PESQ from 1.56 to 1.59, double-talk SDR from −9.26 to −8.86 dB, and far-end single-talk ERLE from 2.21 to 2.83 dB. For STWS, using R or R_m instead of X raises double-talk PESQ from 1.57 to 1.75 and 1.74, and increases ERLE from 9.93 dB to 16.96 and 16.83 dB. WSTWS shows the same input trend: R and R_m improve double-talk PESQ from 1.65 to 1.86 and ERLE from 7.71 dB to 12.66 and 13.36 dB. However, its standalone SDR and ERLE are not consistently higher than those of STWS. Because these comparisons keep the AEC algorithm fixed while changing the reference from X to R or R_m, the larger mismatched gains support the observability-based explanation.

Table 2: AEC results on the mismatched test set.

Test scenarios	DT			ST_NE		ST_FE
Model	PESQ	SDR (dB)	EchoMOS	PESQ	SDR (dB)	ERLE (dB)
Mixture	1.52	−12.0	-	2.28	−36.6	-
PFDAF(Y, X)	1.56	−9.3	2.91	-	-	2.2
PFDAF(Y, R)	1.59	−8.9	2.84	-	-	2.8
$\mathcal{F}_{\text{STWS}}(Y, X)$	1.57	−7.8	3.98	-	-	9.9
$\mathcal{F}_{\text{WSTWS}}(Y, X)$	1.65	−9.0	2.80	-	-	7.7
$\mathcal{F}_{\text{STWS}}(Y, R)$	1.75	−6.8	4.33	-	-	17.0
$\mathcal{F}_{\text{WSTWS}}(Y, R)$	1.86	−7.9	3.36	-	-	12.7
$\mathcal{F}_{\text{STWS}}(Y, R_m)$	1.74	−7.1	4.32	-	-	16.8
$\mathcal{F}_{\text{WSTWS}}(Y, R_m)$	1.86	−8.1	3.46	-	-	13.4
ICCRN	2.05	3.1	4.47	2.92	5.1	36.2
ICCRN_{r}	2.23	3.7	4.58	2.93	5.3	**74.0**
ICCRN_{rl}	2.10	3.4	4.48	2.96	5.1	37.4
$\text{ICCRN}_{\text{rl+fl}}$	2.29	4.2	**4.60**	**2.98**	5.6	65.8
$\text{ICCRN}_{\text{rl+fl+mrl}}$	2.38	4.3	**4.60**	2.93	5.6	70.0
$\text{ICCRN}_{\text{rl+fl+mrl+w}}$	**2.47**	**4.8**	**4.60**	2.97	**5.9**	72.7

The neural post-processing stage further converts these linear estimates into stronger AEC performance. On the matched test set, the full model achieves the best scores on the main PESQ, SDR, and ERLE metrics. Compared with ICCRN, it increases double-talk PESQ from 2.34 to 2.59 and double-talk SDR from 4.09 to 5.14 dB. It also improves near-end single-talk SDR from 5.09 to 5.86 dB and far-end single-talk ERLE from 73.50 to 77.82 dB. These gains show that the neural back-end does not merely replace the linear stage. Instead, it uses the linear AEC outputs as informative residual-echo cues for further enhancement.

The ablation sequence clarifies the role of each input source. Adding the reference-based STWS output improves double-talk PESQ from 2.37 to 2.49 and SDR from 4.05 to 4.71 dB. The far-end-based STWS output then increases ERLE from 73.38 to 76.10 dB. The masked-reference branch raises double-talk SDR from 4.68 to 5.01 dB and ERLE from 76.10 to 77.15 dB. With WSTWS, the full model reaches the best double-talk PESQ (2.59) and SDR (5.14 dB). It also gives the best near-end single-talk SDR (5.86 dB) and far-end single-talk ERLE (77.82 dB). This progression supports the proposed multi-input design. Thus, WSTWS has its strongest effect when used as a neural input, where the back-end can exploit its weighted linear estimate together with the auxiliary and purified references.

Under mismatched nonlinearities, the neural post-processing stage remains effective and provides stronger evidence of generalization. The full model raises double-talk PESQ from 2.05 to 2.47 and double-talk SDR from 3.1 to 4.8 dB compared with ICCRN. These gains are larger than those observed on the matched set. This controlled trend further indicates that the proposed inputs are particu-

larly helpful when the loudspeaker distortion differs from the training condition. The full model also improves near-end single-talk SDR from 5.1 to 5.9 dB and maintains a high far-end single-talk ERLE of 72.7 dB. Although ICCRN_r gives the highest mismatched ERLE (74.0 dB), the full model provides the strongest overall balance across double-talk speech quality, double-talk distortion reduction, and near-end single-talk fidelity. Together, the linear and neural results support the proposed design: the auxiliary reference improves the observability of nonlinear echo, and neural post-processing converts these improved linear estimates into stronger end-to-end AEC performance.

6 Conclusion

This paper addresses the challenge of nonlinear distortions in AEC, commonly caused by low-cost loudspeakers and amplifiers. We propose a novel AEC framework using an auxiliary reference microphone placed near the loudspeaker to capture nonlinear far-end signals. To suppress near-end speech contamination in the reference microphone signal, we propose a short-time Wiener filtering technique followed by real-valued ratio masking to generate a masked, cleaner reference signal. A weighted STWS module further enhances preliminary suppression before neural processing. Evaluation results show that our method significantly outperforms baselines on both matched and mismatched test sets. It achieves the best scores on the main PESQ, SDR, and ERLE metrics, and even greater gains (20.4% in PESQ and 1.7 dB in SDR) under simulated strong unseen nonlinearities in the double-talk scenario. The results demonstrate improved robustness and generalization under these simulated conditions, validating the effectiveness of introducing an auxiliary reference microphone to help AEC and the STWS-based masking strategy.

References

1. J. Benesty, T. Gänsler *et al.*, "Advances in network and acoustic echo cancellation," 2001.
2. E. Hänsler and G. Schmidt, *Acoustic echo and noise control: a practical approach*. John Wiley & Sons, 2005.
3. J. P. Borrallo and M. G. Otero, "On the implementation of a partitioned block frequency domain adaptive filter (pbfdaf) for long acoustic echo cancellation," *Signal Processing*, vol. 27, no. 3, pp. 301–315, 1992.
4. N. Bershad and P. Feintuch, "Analysis of the frequency domain adaptive filter," *Proc. IEEE*, vol. 67, no. 12, pp. 1658–1659, 1979.
5. F. Kuech, E. Mabande, and G. Enzner, "State-space architecture of the partitioned-block-based acoustic echo controller," in *Proc. ICASSP*, 2014, pp. 1295–1299.
6. A. Guérin, G. Faucon, and R. Le Bouquin-Jeannès, "Nonlinear acoustic echo cancellation based on Volterra filters," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 11, no. 6, pp. 672–683, 2004.
7. T. S. Wada and B.-H. Juang, "Enhancement of residual echo for robust acoustic echo cancellation," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 20, no. 1, pp. 175–189, 2011.

8. F. Zhao, J. Liu, and X. Zhang, "SDAEC: Signal Decoupling for Advancing Acoustic Echo Cancellation," in *Proc. Interspeech*, 2024, pp. 172–176.
9. G. Zhang, L. Yu, C. Wang, and J. Wei, "Multi-scale temporal frequency convolutional network with axial attention for speech enhancement," in *Proc. ICASSP*, 2022, pp. 9122–9126.
10. C. Zhang, J. Liu, H. Li, and X. Zhang, "Neural Multi-Channel and Multi-Microphone Acoustic Echo Cancellation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, pp. 2181–2192, 2023.
11. Z.-Q. Wang, S. Cornell, S. Choi *et al.*, "TF-GridNet: Integrating Full- and Sub-Band Modeling for Speech Separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 3221–3236, 2023.
12. N. Howard, A. Park, T. Z. Shabestary, A. Gruenstein *et al.*, "A Neural Acoustic Echo Canceller Optimized Using An Automatic Speech Recognizer and Large Scale Synthetic Data," in *Proc. ICASSP*, 2021, pp. 7128–7132.
13. H. Zhao, N. Li, R. Han, L. Chen *et al.*, "A Deep Hierarchical Fusion Network for Fullband Acoustic Echo Cancellation," in *Proc. ICASSP*, 2022, pp. 9112–9116.
14. S. Panchapagesan, A. Narayanan, T. Z. Shabestary, S. Shao *et al.*, "A Conformer-based Waveform-domain Neural Acoustic Echo Canceller Optimized for ASR Accuracy." Proc. Interspeech, 2022, pp. 2538–2542.
15. K. Sridhar, R. Cutler, A. Saabas, T. Parnamaa *et al.*, "ICASSP 2021 acoustic echo cancellation challenge: Datasets, testing framework, and results," in *Proc. ICASSP*, 2021, pp. 151–155.
16. R. Cutler, A. Saabas, T. Pärnamaa, M. Purin *et al.*, "ICASSP 2023 acoustic echo cancellation challenge," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 675–685, 2024.
17. Z. Chen, X. Xia, S. Sun, Z. Wang *et al.*, "A progressive neural network for acoustic echo cancellation," in *Proc. ICASSP*, 2023, pp. 1–2.
18. Z. Wang, Y. Na, Z. Liu, B. Tian *et al.*, "Weighted recursive least square filter and neural network based residual echo suppression for the AEC-Challenge," in *Proc. ICASSP*, 2021, pp. 141–145.
19. G. Revach, N. Shlezinger, R. J. Van Sloun, and Y. C. Eldar, "Kalmannet: Data-driven kalman filtering," in *Proc. ICASSP*, 2021, pp. 3905–3909.
20. D. Yang, F. Jiang, W. Wu, X. Fang *et al.*, "Low-complexity acoustic echo cancellation with neural Kalman filtering," in *Proc. ICASSP*, 2023, pp. 1–5.
21. Y. Zhang, M. Yu, H. Zhang, D. Yu *et al.*, "Kalmannet: A learnable Kalman filter for acoustic echo cancellation," *CoRR*, 2023.
22. W. Klippel, "Tutorial: Loudspeaker nonlinearities—Causes, parameters, symptoms," *Journal of the Audio Engineering Society*, vol. 54, no. 10, pp. 907–939, 2006.
23. Z.-Q. Wang, G. Wichern, and J. Le Roux, "Convolutive prediction for monaural speech dereverberation and noisy-reverberant speaker separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3476–3490, 2021.
24. F. Zhao and X. Zhang, "Attention-Enhanced Short-Time Wiener Solution for Acoustic Echo Cancellation," in *Proc. ICASSP*, 2025, pp. 1–5.
25. A. Narayanan and D. Wang, "Ideal ratio mask estimation using deep neural networks for robust speech recognition," in *Proc. ICASSP*, 2013, pp. 7092–7096.
26. J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
27. K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on imagenet classification," in *Proc. ICCV*, 2015, pp. 1026–1034.

28. J. Liu and X. Zhang, "ICCRN: Inplace Cepstral Convolutional Recurrent Neural Network for Monaural Speech Enhancement," in *Proc. ICASSP*, 2023, pp. 1–5.
29. A. Li, C. Zheng, R. Peng, and X. Li, "On the importance of power compression and phase estimation in monaural speech dereverberation," *JASA Express Letters*, vol. 1, no. 1, 2021.
30. Y. Sun, L. Yang, H. Zhu, and J. Hao, "Funnel Deep Complex U-Net for Phase-Aware Speech Enhancement," in *Proc. Interspeech*, 2021, pp. 161–165.
31. Y. Luo and N. Mesgarani, "Conv-TasNet: Surpassing Ideal Time-Frequency Magnitude Masking for Speech Separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 27, no. 8, pp. 1256–1266, 2019.
32. J. B. Allen and D. A. Berkley, "Image method for efficiently simulating small-room acoustics," *The Journal of the Acoustical Society of America*, vol. 65, no. 4, pp. 943–950, 1979.
33. C. Zhang, J. Liu, and X. Zhang, "A Complex Spectral Mapping with Inplace Convolution Recurrent Neural Networks For Acoustic Echo Cancellation," in *Proc. ICASSP*, 2022, pp. 751–755.
34. A. Stenger and W. Kellermann, "Adaptation of a memoryless preprocessor for nonlinear acoustic echo cancelling," *Signal Processing*, pp. 1747–1760, 2000.
35. B. S. Nollett and D. L. Jones, "Nonlinear echo cancellation for hands-free speakerphones," *Proc. NSIP'97*, pp. 8–10, 1997.
36. G. Enzner, H. Buchner, A. Favrot, and F. Kuech, "Acoustic echo control," in *Academic press library in signal processing*. Elsevier, 2014, vol. 4, pp. 807–877.
37. A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, "Perceptual evaluation of speech quality (PESQ)-a new method for speech quality assessment of telephone networks and codecs," in *Proc. ICASSP*, 2001, pp. 749–752.
38. E. Vincent, R. Gribonval, and C. Févotte, "Performance measurement in blind audio source separation," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 14, no. 4, pp. 1462–1469, 2006.
39. M. Purin, S. Sootla, M. Sponza, A. Saabas, and R. Cutler, "Aecmos: A speech quality assessment metric for echo impairment," in *Proc. ICASSP*, 2022, pp. 901–905.